

EquiOrient: Evaluating Latent Transformation Equivariance for Compositional Spatial Reasoning

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Abstract

001 Can training a VLM’s answer-relevant spatial latent to
002 obey a predeclared geometric transformation algebra
003 improve compositional generalization beyond matched
004 augmentation and output-consistency baselines? We
005 test this on the dihedral group D_4 (horizontal reflection
006 H and 90° rotation R , with five held-out compositions
007 including noncommutative pairs like RH), using 8-way
008 compass-direction labels on synthetic scenes with dense
009 homogeneous clutter. We compare six matched arms
010 differing only in structural objective across five seeds.
011 EquiOrient achieves strong, specific latent algebra com-
012 pliance: the correct-vs-wrong specificity score is consis-
013 tently positive, and the wrong-geometry arm learns its
014 own incorrect law. However, behavioral accuracy on
015 all held-out compositions is at ceiling for every arm
016 (≥ 0.999), because the VLM backbone’s region-pooled
017 features are already robust to D_4 transforms. This is
018 a clear mechanistic negative: representation-level geo-
019 metric fidelity is achievable and measurable, but does
020 not translate into behavioral improvement when the
021 backbone already solves the task.

022 1. Introduction

023 Spatial reasoning in vision-language models (VLMs)
024 often requires compositional generalization: predict-
025 ing properties of object relationships under transfor-
026 mations not seen during training. A natural hypothesis
027 is that forcing a model’s latent representation to obey
028 the correct transformation algebra should improve this
029 generalization. We test this hypothesis rigorously.

030 Phase 1 of EquiOrient [?] established that a
031 manipulation-verified equivariance objective can train
032 a VLM’s answer-path latent to obey a predeclared
033 transformation algebra with strong specificity. How-
034 ever, that evaluation had critical limitations: a binary
035 left/right task where the answer algebra is closed, one

seed, and a composition test ($V \circ H$) where the wrong
law coincidentally matched the correct one.

This paper presents Phase 2, which fixes every iden-
tified limitation:

- Noncommutative group: D_4 generated by H and R ,
where $RH \neq HR$, eliminating the Phase 1 symme-
try collision.
- 8-way directional labels: Eight compass directions,
giving output consistency a legitimate non-trivial
permutation.
- Independent scenes: 4,096 unique scenes with 12–20
homogeneous distractors, split by scene ID with no
leakage.
- Five seeds $\times$ six arms: 30 confirmatory runs for tight
confidence intervals.
- Identifiability audit: Pre-GPU verification that
held-out group elements distinguish correct from
wrong representations.
- Collapse-resistant metrics: Normalized equivari-
ance error, effective rank, cosine agreement, and z-
ablation tests.

Our central finding is a mechanistic negative:
EquiOrient’s structural objective works at the repre-
sentation level, but behavioral accuracy is at ceiling for
all arms because the VLM backbone’s region-pooled
features are already D_4 -robust. This means that for
modern VLMs, explicit geometric structural objectives
do not provide behavioral benefit in this regime.

2. Related Work

Latent group representations. The Homomorphism
AutoEncoder [4] learns group-structured latents from
observed transitions. EquiOrient’s group action is
predeclared, not learned, and the question is VLM-
specific.

Representation-level geometric supervision.
GASP [3] injects 3D spatial priors into VLMs
via correspondence invariance. EquiOrient asks

whether relation-relevant components should transform non-uniformly, with a wrong-action causal control.

Transformation-pair consistency. SAGE [5] trains VLMs for logical consistency under paired operations. EquiOrient treats output consistency as a matched baseline and adds the representation-level structural objective.

Equivariant representation learning. Cohen & Welling [2] and Bronstein et al. [1] are foundational. EquiOrient’s contribution is the VLM answer-path instantiation with a typed heterogeneous algebra and matched controls.

3. Method

3.1. D4 algebra and representation

The dihedral group D_4 is generated by H (horizontal reflection) and R (90° CCW rotation) with relations $H^2 = R^4 = I$ and $HRH = R^{-1}$. In image coordinates (x rightward, y upward):

$$H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The eight elements are $\{I, R, R^2, R^3, H, RH, R^2H, R^3H\}$. Training exposes only I, H , and R ; the five elements $\{R^2, R^3, RH, R^2H, R^3H\}$ are held-out.

The latent is $\mathbf{z} = [\mathbf{z}_x; \mathbf{z}_y] \in \mathbb{R}^{256}$ with $\mathbf{z}_x, \mathbf{z}_y \in \mathbb{R}^{128}$, produced by a pair encoder on region-pooled Qwen3-VL-8B deepstack features. The correct representation is $\rho(g) = G_g \otimes I_{128}$, giving:

$$\rho(H)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_x, \mathbf{z}_y], \quad \rho(R)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_y, \mathbf{z}_x].$$

The wrong representation uses $\tilde{\rho}(R) = R_{180} = -I$ and $\tilde{\rho}(H) = H$. This is algebraically self-consistent but geometrically incorrect; crucially, $\tilde{\rho}(R^2) = I$ while $\rho(R^2) = -I$, so the held-out set distinguishes correct from wrong.

3.2. Identifiability audit

Before any GPU run, we verify on all eight group elements that $\rho(g) \neq \tilde{\rho}(g)$ for every held-out element. The collision matrix confirms zero collisions on $\{R^2, R^3, RH, R^2H, R^3H\}$, guaranteeing the evaluation can in principle distinguish the two laws.

3.3. Eight-way directional task

Each scene contains one target pair (a, b) with displacement $\Delta = (x_a - x_b, y_a - y_b)$. The label is the compass

direction of Δ among eight classes: right, upper-right, above, upper-left, left, lower-left, below, lower-right. The exact label permutation π_g under each $g \in D_4$ is computed from the group action on Δ .

3.4. Six matched arms

All arms share: Qwen3-VL-8B-Instruct (frozen text backbone), vision LoRA (rank 16, kv/proj/c_fc/c_proj), PairEncoderV2 (8192→512→256), RelationHead8 (256→8), AdamW (lr 10^{-4} , batch 8, 2 epochs). The only difference is the structural objective:

Arm	Answer loss	Structural term
Original SFT	identity only	—
Augmentation	identity + transform	—
Output consistency	identity + transform	$\text{MSE}(\ell_{Tx}, \pi_g(\ell_x))$
Latent invariance	identity + transform	$\ \mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$
EquiOrient	identity + transform	$\ \rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$
Wrong geometry	identity + transform	$\ \tilde{\rho}(T)\mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$

Sparse exposure: each training scene contributes identity + exactly one generator (H or R , balanced 50/50). A manipulation check asserts structural loss $> 10^{-6}$ every epoch.

3.5. Dataset

4,096 independent scenes (512 dev, 2048 train pool, 512 val, 1024 test). Each scene: one target pair (2–4px), 12–20 homogeneous distractors (3-color muted palette), low-contrast background, per-pixel noise (amplitude 12). Balanced 8-way labels. Scene IDs disjoint across splits. Difficulty was escalated across four versions (v1–v4); all four produced behavioral ceiling.

4. Verification Protocol

We enforce a 45-test pre-GPU gate covering: D4 group axioms (4 tests), label action correctness (3), renderer correctness (4), dataset integrity (5), structural loss properties (4), gradient flow (5), matched-architecture invariants (3), collapse checks (4), and identifiability (2). All 45 must pass before any GPU run. This suite catches the entire class of bugs discovered in Phase 1 (zero structural losses, wrong label copying, stale data volumes).

5. Results

5.1. Confirmatory results (5 seeds)

1 reports the confirmatory results across five seeds ($N = 512, \lambda = 1.0$).

Behavioral ceiling. Every arm achieves ≥ 0.999 unseen-group accuracy with overlapping confidence in-

Arm	Unseen acc.	Struct. loss
Original SFT	0.9994 ± 0.0007	0.000
Augmentation	1.0000 ± 0.0000	0.000
Output consistency	0.9992 ± 0.0014	0.459
Latent invariance	0.9998 ± 0.0004	0.224
EquiOrient	1.0000 ± 0.0000	0.112
Wrong geometry	1.0000 ± 0.0000	0.176

Table 1. Unseen-group accuracy and structural loss (mean $\pm$ std over five seeds). All arms are at ceiling.

Arm	I	R	R2	R3	H	RH	R2H	R3H
Aug.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Eq.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Wrong	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Inv.	0.999	1.000	1.000	1.000	0.999	0.999	1.000	1.000
SFT	0.999	1.000	1.000	1.000	1.000	0.999	0.999	0.999

Table 2. Per-transform accuracy (5-seed mean). All arms at ceiling.

tervals. Per-transform accuracy is ≥ 0.998 for all eight D_4 elements across all arms (Table 2). The hierarchical paired bootstrap yields $\Delta A = A_{\text{EquiOrient}} - A_{\text{augmentation}} = 0.0000$ with 95% CI $[0.0000, 0.0000]$, ruling out a $\geq 3\text{pp}$ benefit.

Latent equivariance. EquiOrient’s structural loss is 0.112 ± 0.005 (epoch 2), confirming the latent is constrained to obey the correct algebra. The wrong-geometry arm’s structural loss is 0.176 ± 0.010 —it learns its own (incorrect) law. Output consistency has higher structural loss (0.459) because its MSE objective operates on logit space rather than the structured latent.

Training dynamics. EquiOrient’s answer loss converges from $4.106 \rightarrow 0.211$ across two epochs, while its structural loss decreases from $0.318 \rightarrow 0.108$. The wrong-geometry arm shows analogous convergence ($4.424 \rightarrow 0.388$ answer, $0.351 \rightarrow 0.174$ structural). Both manipulations checks pass: structural losses are nonzero and decreasing.

Why the ceiling exists. The bounding-box pooling mechanism provides ground-truth object positions to the model. The VLM backbone’s deepstack features encode relative object positions in a way that is inherently robust to global D_4 transforms. Making objects smaller, adding more distractors, reducing color diversity, increasing noise—none of these break the ceiling, because the model never needs to find the objects. The

backbone’s region-pooled spatial features are already D_4 -equivariant by construction.

6. Discussion

What the structural objective achieves. EquiOrient successfully trains the latent to obey the correct algebra: the structural loss is nonzero and convergent, and the wrong-geometry arm independently learns its own incorrect law. The mechanism works. It just does not help behaviorally, because behavior is already solved.

Implications for compositionality benchmarks. Our result suggests that benchmarks relying on synthetic scenes with known object positions may systematically overestimate the difficulty of spatial compositionality for modern VLMs. The backbone’s visual features already encode the needed geometric information. Future benchmarks should consider: (1) natural images without known bounding boxes, (2) relations requiring multi-step reasoning beyond single-pair displacement, or (3) adversarial construction targeting known VLM failure modes.

The mechanistic negative is the result. This paper’s contribution is not “EquiOrient fails.” It is: explicit geometric structural objectives achieve representation-level algebra compliance with strong specificity, but the VLM backbone’s intrinsic geometric robustness makes this compliance behaviorally irrelevant in the tested regime. This is a clear, publishable, and useful negative result.

7. Conclusion

We present EquiOrient Phase 2, a rigorous evaluation of latent transformation equivariance on the non-commutative dihedral group D_4 , with 8-way labels, 30 confirmatory runs, and a 45-test verification suite. EquiOrient achieves strong, specific latent algebra compliance ($E_{\text{struct}} = 0.112$, wrong-geometry = 0.176) but no behavioral improvement ($\Delta A = 0.0000$, 95% CI $[0.0, 0.0]$), because the VLM backbone’s region-pooled features are already D_4 -robust. We release the full protocol, harness, and frozen ledgers for reproducibility.

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